

The Application Layer

What does the application layer do?

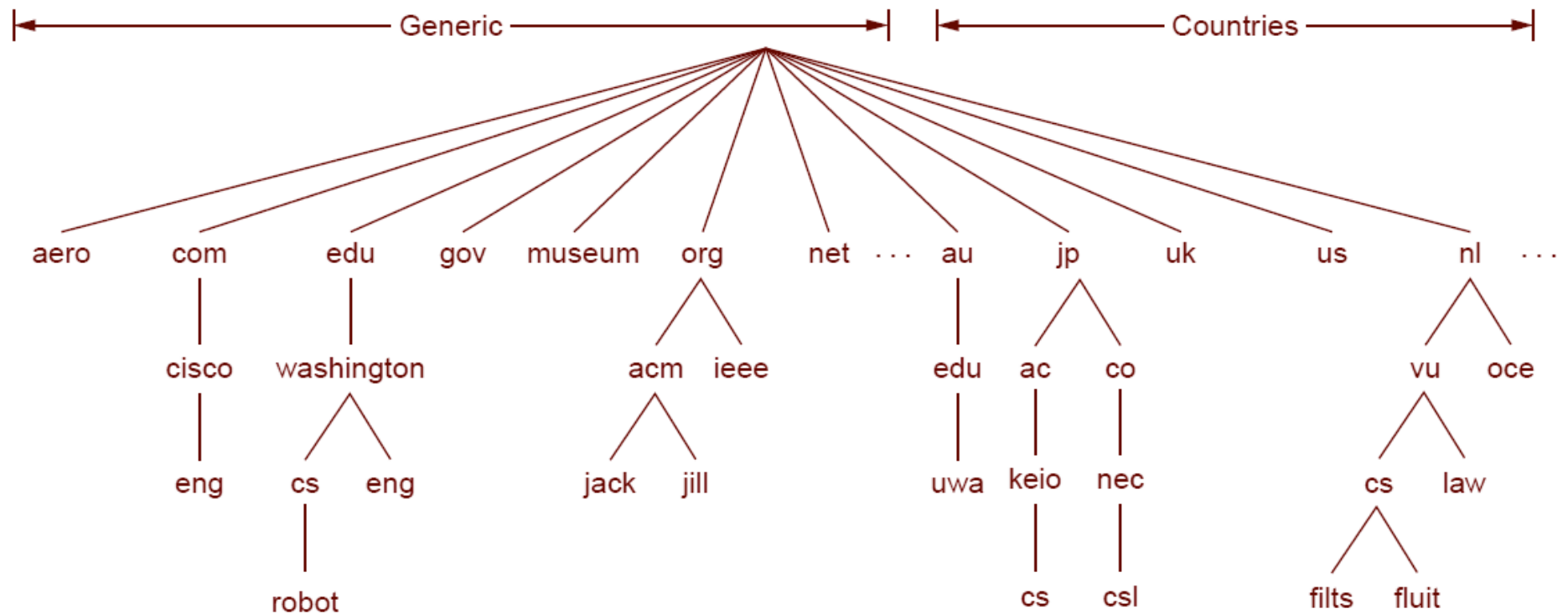
- ◆ Provides services to *user*
 - ◆ Domain Name System (*DNS*)
 - ◆ File Transfer Protocol (*FTP*)
 - ◆ Simple Mail Transfer Protocol (*SMTP*)
 - ◆ Hypertext Transfer Protocol (*HTTP*)
 - ◆ ...

Domain Name System (*DNS*)

- ◆ Support service for several protocols
- ◆ Names are *easier* for humans to remember than IP addresses
- ◆ Names help with *portability* even when IP address changes
- ◆ DNS resolves human-readable names to IP addresses

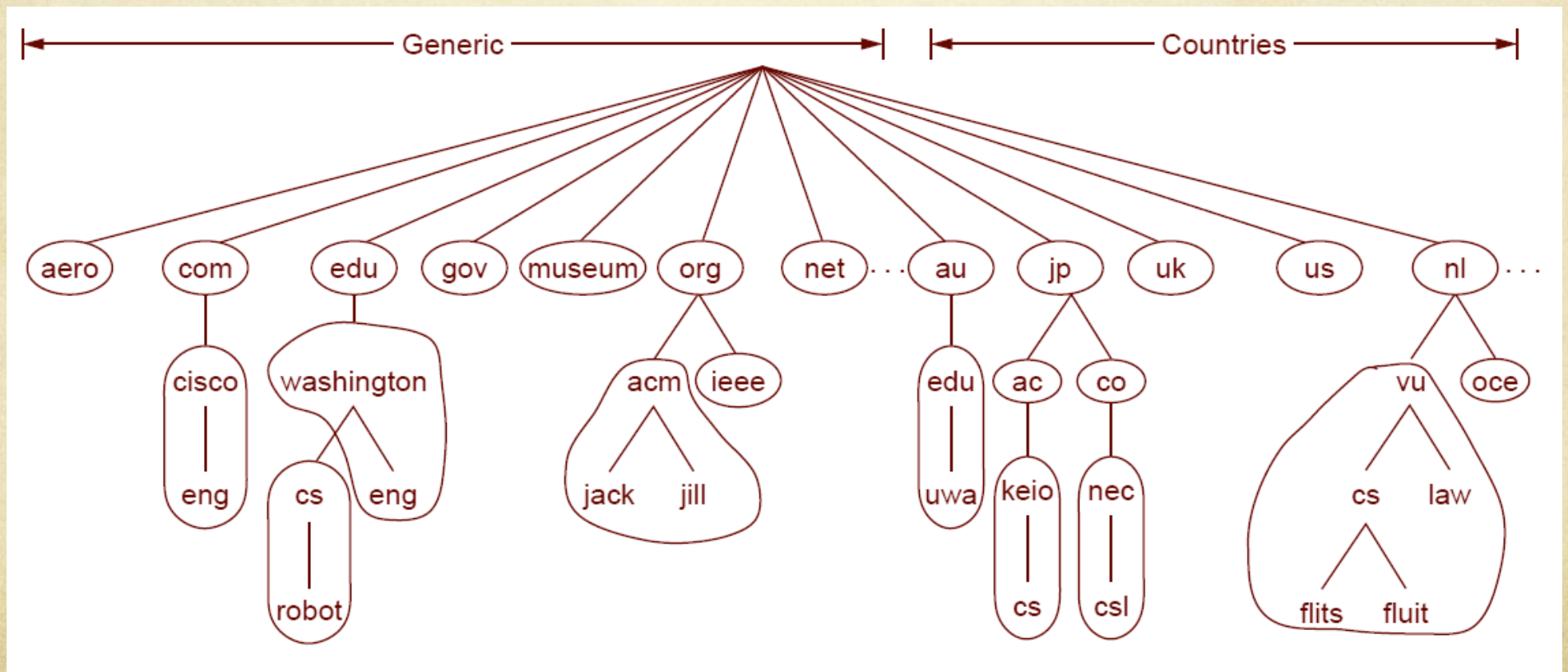
Domain Name System (*DNS*)

- ◆ Supports hierarchical, domain-based naming system



Domain Name System (*DNS*)

- ◆ *Name servers* → contain data for portions of name space called *zones*



Domain Name System (*DNS*)

- ◆ Finding IP address for given hostname → *resolution*
 - ◆ Computer requests *local* name server to resolve
 - ◆ Local name server asks *root* name server
 - ◆ Root returns name server for *lower* zone
 - ◆ Continue down zones until name server can answer

Domain Name System (*DNS*)

◆ Example

